01 - Follow a message through a network

Read, predict and model

Name _____ Class _____ Date _____

Separate local forwarding, routing and application responsibilities.

Textbook: networks

Before starting: Read Network foundations before Activity 01. Activities 01-03 develop Unit 1 reasoning; Activity 04 is a Year 12 Unit 3 addressing bridge and requires binary/prefix knowledge. Use supplied fictional addresses and logs; no scanning or live configuration is needed.

A classroom laptop requests a page from a remote server. Simplify the path as laptop -> access point -> switch -> router -> remote network -> server. The access point connects wireless clients; the switch forwards local frames using link-layer addresses; the router forwards IP packets between networks.

Responsibility	Worked decision
DNS	Resolve a hostname to an IP address; not the page itself
IP / router	Address and forward packets across networks
TCP	Provide ordered reliable byte-stream transport
HTTPS	HTTP protected by TLS; confidentiality and authentication depend on correct verification

These are responsibilities, not a claim that every real network uses this exact arrangement.

Equipment-free route

Use role cards and paper message envelopes. Label the destination and move the envelope along the drawn path; no live traffic capture is required.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

01 - Follow a message through a network

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent trace

Two wired computers are on the same subnet and switch. Draw their simplified message path. Explain why sending this local traffic does not require the internet router to forward it.

B. Diagnose from evidence

A cached page opens, but a new hostname does not resolve. Explain why the cached page is insufficient evidence that DNS works. Name one extra supplied observation you would request before choosing a cause.

Exit

Distinguish the switch forwarding a local frame from the router forwarding a packet between networks.

02 - Compare transport using a supplied trace

Read, predict and model

Name _____ Class _____ Date _____

Explain reliability and latency trade-offs without treating protocols as guarantees.

Textbook: networks

For a simplified teaching trace, a TCP sender transmits segments carrying bytes starting at sequence 100 and 104, four bytes each. The receiver gets the first segment and acknowledges 104, the next expected byte. If the second segment is lost, TCP can retransmit it and the receiver can acknowledge 108. Actual traces include further flags, timing and congestion behaviour.

Application need	Worked reasoning
File transfer	Ordered delivery and recovery matter; TCP is a suitable transport choice
Live audio	Late data can lose value; an application may prefer UDP with its own handling

UDP itself does not promise delivery or ordering. TCP reliability does not guarantee low delay or that an application succeeds.

Equipment-free route

Write sequence numbers on paper strips and simulate loss by withholding a strip. The receiver writes the next expected byte number; this is a teaching model, not a full protocol implementation.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

02 - Compare transport using a supplied trace

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent trace

Repeat the simplified example with first sequence 200 and two segments of 6 bytes. State the next expected byte after the first and after both. If the second is missing, identify what still needs recovery.

B. Choose with reasons

Choose and justify a transport for downloading a marked report and for a live voice stream. State one cost or limitation of each choice. Explain why an application can add reliability over UDP.

Exit

Is a missing packet in a single supplied trace proof of a permanently faulty cable? Give an alternative cause and the next evidence needed.

03 - Design against capacity and access requirements

Read, predict and model

Name _____ Class _____ Date _____

Calculate a lower-bound transfer time and justify a restricted network design.

Textbook: network-design

A fictional school transfers a 120 MB file over a 40 Mbit/s link. Using decimal units, 120 MB = 960 Mbit, so the ideal lower bound is 960 / 40 = 24 seconds. Protocol overhead, contention and storage performance can increase actual time. MB is bytes; Mbit is bits.

The school separates a student segment and a staff segment. A proposed rule permits student access to a named learning server but denies access to a staff file service. The design requires both an allowed-case test and a denied-case test; drawing two colours alone does not enforce isolation.

Equipment-free route

Draw the topology and apply written permit/deny cards to supplied requests. Calculate transfer time on paper. Optional network pods may later test the same rules in an authorised isolated lab.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

03 - Design against capacity and access requirements

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent calculation

A 75 MB file uses a 20 Mbit/s link. Calculate the ideal minimum seconds and name two reasons an observed transfer might be slower. State all unit conversions.

B. Design and test

Draw a small network for 12 student devices, 2 staff devices and a learning server. Mark switches/access points, the routing boundary and the policy enforcement point. Write one permit rule and one deny rule, plus expected outcomes for two tests. Do not configure a live system.

Exit

Why does successfully opening the learning server fail to prove that staff files are protected?

04 - Year 12 bridge: allocate a small subnet

Read, predict and model

Name _____ Class _____ Date _____

Calculate an IPv4 subnet range and include every required interface.

Textbook: unit3-addressing

Year 12 bridge: /27 leaves 5 host bits. It contains 2^5 = 32 addresses. For an ordinary IPv4 LAN subnet with network and broadcast addresses reserved, 30 are usable. In 192.0.2.64/27, network .64, usable .65-.94 and broadcast .95. 192.0.2.0/24 is documentation space used here as a fictional example.

A LAN needs 24 workstation interfaces plus 1 router interface: 25 usable addresses. /27 is the smallest conventional subnet that fits; /28 offers only 14 usable. Other special subnet types have different rules and are outside this activity.

Equipment-free route

Use a 0-255 number line divided into blocks of 32 or 64. Assign interface cards only to usable addresses; no physical router or subnet calculator is needed.

Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

04 - Year 12 bridge: allocate a small subnet

Independent practice and evidence

Name _____ Class _____ Date _____

A. Independent range

For 192.0.2.96/27, give network, broadcast, first and last usable addresses, subnet mask and usable count. Show the block-size method.

B. Changed requirement

The LAN grows to 30 workstations plus 1 router interface. Does /27 still fit? Give the smallest conventional prefix that fits. Allocate it starting at 192.0.2.128 and state its full address range.

Exit

Two adjacent /27 subnets start at .64 and .96. Does .95 belong to the second subnet? Explain its role.